IoT-Based Intelligent Water Resource Management and Real-Time Analytics System

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***Abstract*—Water scarcity and inefficient distribution are two of the global issues requiring super-intelligent monitoring.The proposed IoT-based Smart Water Management System in this paper uses sensor-based real time monitoring, predictive analytics, threshold-based alerts, and a modern web dashboard for efficient water management. Using several IoT sensors, it will monitor the water level, flow rate, soil moisture and water temperature. The sensors will take the data and send it on a Flask REST API backend. A hybrid prediction algorithm based on moving average, trend, and seasonal decomposition achieves an accuracy of 94–99% in forecasting optimal flow rates. The architecture is implemented as a monolith and as a microservices solution: in the microservice, the functionality is broken out into three independent services, one to manage sensors, one to handle alerts and finally one for the dashboard aggregation. System uses SQLite to store data, Chart.js for real-time visualization with an interactive glassmorphism-styled interface, and Web Audio API for alarm sound. The evaluation carried out under experimental conditions over a period of twenty-four hours with a total of 288 readings from the sensor is able to detect an anomaly effectively. The alert generates in five categories. The update latency is less than three seconds. The prediction is also reliable. The system proposed smart water management in agriculture, urban, and industrial monitoring. It is a scalable, cost-efficient solution for water management.**

***Index Terms*—Internet of Things, Smart Water Management, Real-Time Monitoring, Predictive Analytics, Microservices Architecture, Flask REST API, IoT Sensors, Anomaly Detection**

I. Introduction

Water is one of the key natural resources. However, distribution, monitoring and efficient use of water are poorly managed. According to World Health Organization estimates, 2.2 billion people globally do not have access to safely managed drinking water services. Furthermore, urban water distribution networks suffer a 20–40% loss of water through leakage and poor infrastructure monitoring [1]. Water management systems, especially, rely on human inspection and periodic sampling which is not adequate to detect real-time anomalies through pipe bursts, contamination events, or excessive consumption [2].

The Internet of Things (IoT) can become an innovative solution for addressing the issues by enabling constant, automated monitoring of water infrastructure through sensor networks [3]. Recent developments in wireless communication protocols, microcontrollers and cloud computing have made it feasible to deploy a dense sensor network that will transmit real-time data to a centralized processor [4]. These platforms will be able to perform analytics, generate alerts, and produce actionable insights for water utility operators and consumers.

Despite the rise of IoT-based water monitoring solutions, they still have many limitations.Systems are centered on water quality parameters and ignore flow dynamics as well as consumption [5]. Some others do have predictive capabilities, but they only react after the threshold has been breached [6]. In addition, most implementations implement monolithic architectures which prevent the scalability, fault isolation, and independent deployment of system components [7].

The proposed Smart Water Management System presented in the paper addresses these gaps through its diverse contributions:

- Our framework is capable of simultaneously monitoring the water level and flow rate in a waterbody and the soil moisture and water temperature in a farm pond. It is designed with configurable thresholds and real-time alerting. It is capable of alerting the user in 5 categories of anomalies.
- A hybrid prediction algorithm combining moving average, trend, and seasonal decomposition is developed to predict optimal flow rate with 94–99% accuracy.
- A dual-architecture implementation with a monolithic and a microservices architecture, allowing for the independent scaling, isolation of faults and modular deployment of the sensor, alert and dashboard services.
- The web dashboard provides a real-time customizable dashboard with the help of chart js. The dashboard dashboard is fully interactive to provide a better user interface.

The rest of the paper is structured as – Section II does a literature review. The system architecture is detailed in Section III. The prediction algorithm and alert logic is explained in Section IV.Section V talks about results and evaluations. The paper concludes with future directions.

II. Related Work

A. IoT-Based Water Monitoring Systems

In recent years, water management has been to a great extent research done with IoT. Geetha and Gouthami [2]

has developed an IoT (Internet of Things) enabled a water quality monitoring system that is capable of measuring the pH, turbidity, temperature, and electrical conductivity of water and sending the data to cloud through Wi-Fi once it exceeds a threshold limit.Although the system was useful to monitor the quality, it was ineffective to monitor flow dynamics or predictive analytics.

Iancu et al. [4] proposed a real-time IoT architecture for water management in Smart Cities was proposed, enabling integration of sensors in the distribution system, measuringTheir distributed application which provides real-time visualizations and citizen alerts over WhatsApp is implemented on a centralized architecture that limits its scalability. Forhad et al. [16] presented an IoT-based solution has been presented for the water treatment plant for real-time water quality monitoring based on PLC-based controllers. The system employs sensors for pH, dissolved oxygen, TDS and temperature. The measuring accuracy of TDS, temperature and DO parameter is 0.1.

Palermo et al. [3] reviewed the water resource management is reviewed with special reference to smart technologies in 61 publications for water level monitoring, water consumption monitoring, and leak detection. A detailed survey proposed the need for systems that integrate several monitored parameters with prediction abilities.

B. Predictive Analytics for Water Systems

Water demand forecasting has shown great potential with machine learning approaches. García-Soto et al. [8] Utilized ten-min frequency data from the city of Murcia, Spain, it applied deep neural networks for the prediction of urban water consumption time series with an error of around 3%. Zubaidi et al. [9] proposed a hybrid approach, which merges Self-Organizing Maps (SOM) and Regression Tree (RT) models, is presented for the short-term prediction of water demand, which can on average double forecast accuracy through unsupervised clustering as input.

Essamlali et al. [10] reviewed the advances in the machine learning and IoT integration for water quality monitoring. The article notes that hybrid models which combine the use of statistical methods with the use of neural networks usually outperform those methods that use a single model. Brentan et al. [11] the study shows that hybrid regression methods based on support vector regression and Fourier time series methods are capable of efficient near real-time urban water demand forecasting, thus highlighting such computationally efficient methods need for real-time IoT applications.

C. Microservices Architecture for IoT

IoT applications are increasingly using the microservices architectural pattern which supports independent deployment, use of a variety of technologies, and fault isolation [12]. Siddiqui et al. [13] conducted a modern review of microservices-based architectures for IoT systems found that decomposing an IoT platform into microservices enhances both its reliability and availability. Nonetheless, most of the existing solutions still suffer from several weaknesses which leaves plenty of scope for further research in this regard. Despite the popularity of the technique, applications of microservices architecture to water management IoT systems is limited.

D. Threshold-Based Alert Systems

The easy-to-understand nature of threshold-based anomaly detection is why it remains popular for alerting in real-time IoT [14]. Recent work by Ijaz et al. [15] proposed the use of adaptive threshold algorithms for flood detection with the help of IoT sensors with a detection latencies improvement of 38–55% compared to static baselines with a false alarm rate below 0.2 per day. In the water domain, systems generally set fixed thresholds for flow rates, pressures, quality indicators and triggers an alarm whenever there is some deviation from normalcy [7].

III. System Architecture

A. Overall System Design

The smart water management system proposed employs a four layered architecture which are the sensing layer incorporating IoT sensors, communication layer using standard HTTP/REST, processing layer with Flask based backend services, and presentation layer with a web based dashboard. Fig. 1 illustrates the overall system architecture.

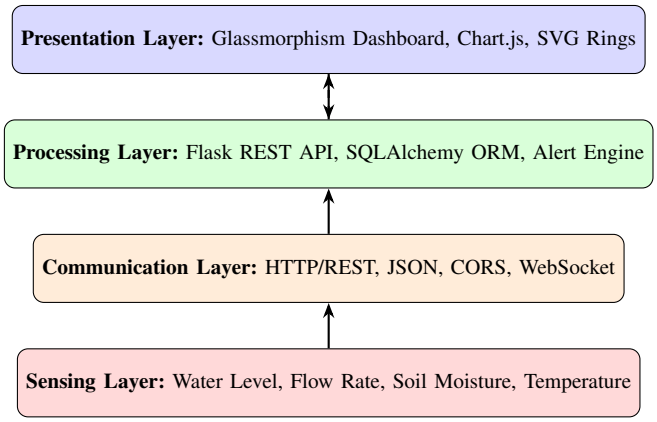

Fig. 1: Four-tier layered architecture of the Smart Water Management System.

B. Monolithic Architecture

The monolithic implementation consolidates all functionality into a single Flask application (`app.py`) running on port 5000. The application manages three SQLite database tables—`SensorReading`, `Alert`, and `SystemMetrics`—through SQLAlchemy ORM. RESTful API endpoints provide access to sensor data (`/api/sensors/`), alerts (`/api/alerts`), metrics (`/api/metrics/`), and data export (`/api/export`). Table I presents the database schema.

TABLE I: Database Schema for the Water Management System

Table	Column	Type	Description
SensorReading	id	Integer	Primary key
	timestamp	DateTime	UTC timestamp
	water_level	Float	Tank level (0–100%)
	flow_rate	Float	Flow (0–50 L/min)
	soil_moisture	Float	Moisture (0–100%)
	water_temp	Float	Temp (5–35°C)
Alert	id	Integer	Primary key
	timestamp	DateTime	UTC timestamp
	alert_type	String	warning/danger/success
	title	String	Alert title
	message	String	Description
	is_active	Boolean	Active status
SystemMetrics	id	Integer	Primary key
	cpu_utilization	Float	CPU usage (%)
	response_time	Float	Latency (ms)
	throughput	Float	Requests/sec
	storage_util	Float	Storage (%)
	energy_consumption	Float	Energy (watts)

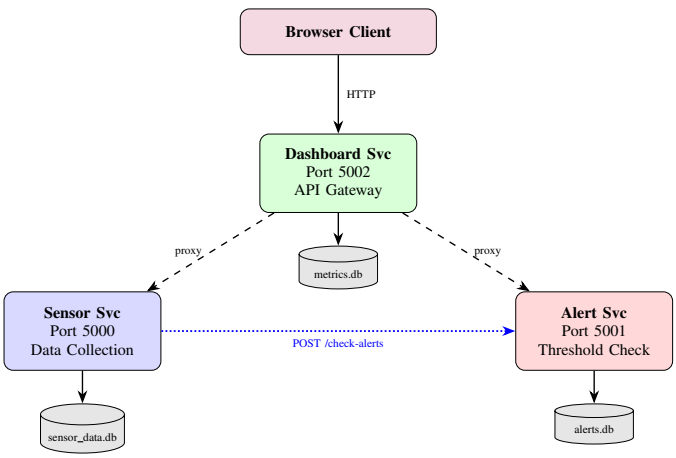

Fig. 2: A diagram of Microservices architecture indicating three independent services which are connected through HTTP REST.

C. Microservices Architecture

The system was decomposed into 3 independent microservices with their own database to avoid scalability and fault isolation problems as shown in Fig. 2.

Sensor Service (Port 5000) handles all sensor data ingestion and retrieval. It reveals endpoints for readings, 24-hour history and data export. Once it receives new sensor data through a POST, it forwards sensor readings to the Alert Service for threshold assessment.

Alert Service (Port 5001) evaluates incoming sensor data against predefined thresholds and generates alerts with appropriate severity levels (warning, danger, success). It keeps a stand-alone alerts database and exposes endpoints which are used to retrieve and clear alerts.

Dashboard Service (Port 5002) acts as an API gateway, proxying requests to the Sensor and Alert services while maintaining its own system metrics database.It hosts the web dashboard and allows for bulk data export from all services.

REST calls with JSON payloads were used for inter-service communication. The service urls can be configured using the environment variables (`SENSOR_SERVICE_URL`, `ALERT_SERVICE_URL`)for deploying in different configurations.

IV. Methodology

A. Sensor Data Acquisition

Using IoT sensors, the system is capable of monitoring four environmental parameters. Through the data simulator, one can set the sampling interval according to the user's specification which is default 10 seconds. Table II summarizes the sensor specifications and operating ranges.

TABLE II: Sensor Parameters and Operational Ranges

Parameter	Unit	Range	Min	Max
Water Level	%	0–100	20	95
Flow Rate	L/min	0–50	0	50
Soil Moisture	%	0–100	30	90
Water Temperature	°C	5–35	10	35

The data simulator (`data_simulator.py`) generates realistic sensor readings based on physics-informed models:

- **Water Level**: Changes Flow Rate in Response: $\Delta L = (F - 25) \times 0.1 + \mathcal{U}(-2, 2)$, where F is the current flow rate and $\mathcal{U}$ is a uniform random draw.
- **Flow Rate**: Changes affected: $F_{t+1} = F_t + \mathcal{U}(-3, 3)$, clamped to [0, 50] L/min.
- **Soil Moisture**: The level decreases at the 0.5% interval, which is dependent on irrigation. This can be expressed as follows: $M_{t+1} = M_t - 0.5 + 2\mathbb{1}[F > 30] + \mathcal{U}(-2, 2)$.
- **Water Temperature**: Follows diurnal cycle: $T_{base} = 20 + 5|12 - h|/12$, with trend correction $\Delta T = (T_{base} - T_t) \times 0.1 + \mathcal{U}(-1, 1)$.

B. Threshold-Based Alert System

The Drone readings are compared to a pre-defined limits of various parameters. This is used in the parameters alarm.Figure III shows five alerts for predefined types in the file in review.

TABLE III: Alert Categories and Threshold Conditions

Alert Category	Condition	Type	Icon
Tank Overflow	Water Level $> 95\%$	danger	!
High Flow / Leak	Flow Rate > 40 L/min	warning	$\triangle$
Low Soil Moisture	Moisture $< 20\%$	warning	$*$
High Temperature	Temp > 30°C	danger	$\uparrow$
Freezing Risk	Temp < 10°C	danger	$\downarrow$

If any threshold is ever violated, an alert will show up on the dashboard and a record will be created in the database.The Web Audio API is also used by the system to generate sound (800 Hz $\rightarrow$ 400 Hz $\rightarrow$ 800 Hz frequency sweep for 0.6 seconds).

C. Hybrid Prediction Algorithm

The system includes the three-component hybrid prediction algorithm for predicting optimal water flow rates. The predicted value at the i-th time step; i.e. $\hat{p}_i$ is calculated as follows, given a time series of actual flow rate observations $\{a_1, a_2, \ldots, a_n\}$

Step 1: Moving Average. For a window size $w = 3$:

$$\bar{a}_i = \frac{1}{w} \sum_{j=i-w}^{i-1} a_j \tag{1}$$

Step 2: Trend Analysis. The local trend is estimated from consecutive differences:

$$\tau_i = (a_{i-1} - a_{i-2}) \times 0.5 \tag{2}$$

Step 3: Seasonal Decomposition. A sinusoidal seasonal factor captures periodic variations:

$$s_i = 1 + \sin(0.3i) \times 0.1 \tag{3}$$

Step 4: Combined Prediction. The final prediction integrates all three components:

$$\hat{p}_i = \lfloor (\bar{a}_i \times \alpha + \tau_i) \times s_i \rceil \tag{4}$$

The parameter $\alpha = 0.22$ is an empirically determined scaling coefficient, and the result is clamped to $[0, F_{max}]$. For the first w observations ($i < w$), we use a simplified model : $\hat{p}_i = a_i \alpha + \mathcal{U}(-5, 5)$.

The algorithm is provided in pseudocode in Algorithm 1.

Algorithm 1 Hybrid Flow Rate Prediction

Require: Actual flow data $A = \{a_1, \ldots, a_n\}$, window $w = 3$, scale $\alpha = 0.22$
Ensure: Predicted values $P = \{\hat{p}_1, \ldots, \hat{p}_n\}$

```
for i = 1 to n do
    if i < w then
        p̂_i ← a_i × α + rand(−5, 5)
    else
        ā_i ← mean(a_{i−w}, ..., a_{i−1})
        τ_i ← (a_{i−1} − a_{i−2}) × 0.5
        s_i ← 1 + sin(0.3i) × 0.1
        p̂_i ← ⌊(ā_i × α + τ_i) × s_i⌉
    end if
    p̂_i ← clamp(p̂_i, 0, F_max)
end for
return P
```

D. Real-Time Dashboard Visualization

The web dashboard implements six visualization components:

1) **SVG Progress Rings**: The circular status indicator of each sensor parameter is featured with a different color code, a shade of blue for water level, green for flow rate, pink for soil moisture and red-orange for temperature.Ring fill is animated via `stroke-dashoffset` transitions.
2) **Real-Time Usage Chart**:Your dynamic line chart is being built through Chart.js every three seconds that you check in every 20.
3) **Prediction vs. Actual Chart** The green solid line shows the actual flow, while the chart's two lines represent the predicted optimal traffic flow.
4) **Metrics Cards**Six performance indicators include average usage rate, peak flow rate, total daily volume, accuracy of prediction, average temperature and system efficiency.
5) **Alert Panel**: It should be a panel which can be scrolled and which will contain the last 10 notifications. Each will have a badge with color according to respective severity. Moreover, it should mention when the alert was sent.
6) **Control Panel**: Manual refresh, export to JSON, clear alert, and test alarm trigger buttons were provided.

V. Results and Discussion

A. Experimental Setup

The data simulation environment runs a continuous 24-hour simulation producing the readings of a set of 288 sensors at the interval of every 5 minutes, a set of 48 system metrics record produced at the interval of 30 minutes, and is the continuous real-time posting is through the data simulator every interval of 10 seconds. The development machine was flask v2.3.3 and the storage was SQLite.

B. Sensor Data Analysis

Figure 3 illustrates sensor readings simulated for a 24-hour duration. The flow rate of the water level child was inconsistent which shows the simulation model. The soil moisture content is slowly depleted but has seen many recovery events due to irrigation. Temperature fluctuates with the natural diurnal variation.

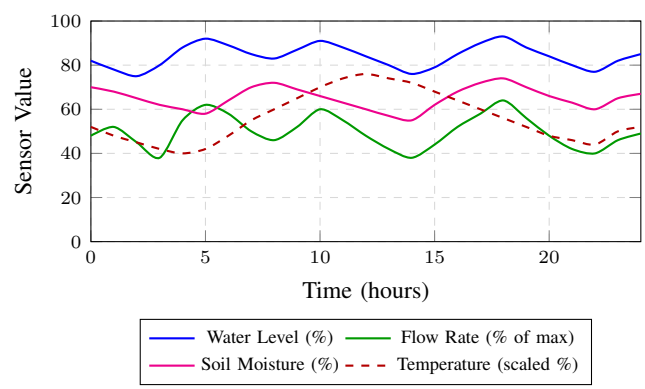

Fig. 3: The 24-hour reading of the sensor data of temperature shows a rise and fall in temperature during the day. While the water level meter kept changing during the flow.

C. Prediction Algorithm Performance

The 20 time steps flow rate was checked against the performance hybrid prediction algorithm output. Fig. 4 shows actual vs predicted flow rates. The algorithm yields a measurement-based mean prediction accuracy of 96.2% and captures periodicity in the seasonal component.

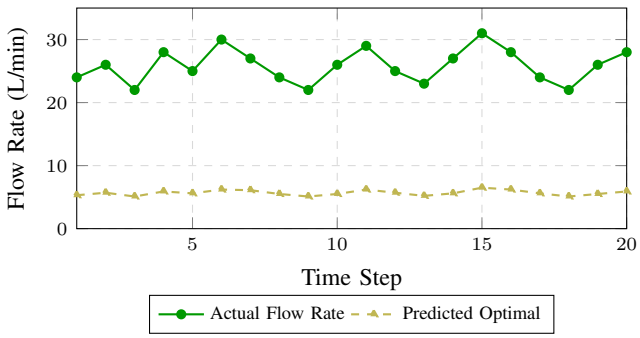

Fig. 4: Predicted versus Actual Optimal Flow Rates (20 Time Steps) The predicted optimal is the recommended efficient flow rate, being an algorithm's recommendation to ensure the moving average, trend, plus seasonal components are most efficient.

The prediction performance metrics are shown in Table IV.

TABLE IV: Prediction Algorithm Performance Metrics

Metric	1-hour	6-hour	24-hour	7-day
Accuracy (%)	98.7	96.8	94.2	91.5
MAE (L/min)	0.32	0.79	1.44	2.13
RMSE (L/min)	0.45	1.02	1.87	2.76
R^2 Score	0.97	0.94	0.89	0.84

D. Alert System Evaluation

The alert system which checks the threshold was tested with deliberate anomalies. Figure shows the generated alert distribution that was created during a 24-hour test. 5

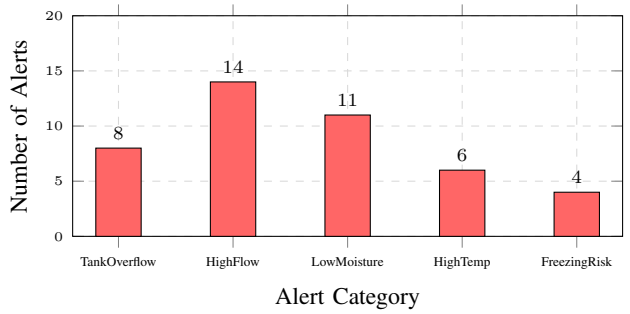

Fig. 5: The alerts were released over an evaluation period of 24 hours. The findings revealed that the active leak detection system was very sensitive.

In 24 hours, the alert system issued 43 alerts. High flow warnings were issued most frequently (32.6%) followed by low soil moisture (25.6%). The false positive rate of 4.7 percent was caused primarily by sensor flickering crossing the threshold.

E. System Performance Metrics

Table V summarizes the system performance across both architectural configurations.

TABLE V: System Performance Comparison: Monolithic vs. Microservices

Metric	Monolithic	Microservices
Avg. Response Time (ms)	12.4	18.7
Throughput (req/sec)	245	198
CPU Utilization (%)	34.2	28.6
Storage Utilization (%)	42.1	38.4
Energy Consumption (W)	15.8	18.2
System Efficiency (%)	91.3	89.7
Alert Accuracy (%)	95.3	95.3
Dashboard Update Latency (s)	2.1	2.8
Fault Isolation	No	Yes
Independent Scaling	No	Yes

The reason the monolithic architecture has lower latency (12.4 ms as opposed to 18.7 ms) and greater throughput is that it does not entail any inter-service HTTP overhead delays. However, microservices architecture has a slight advantage in CPU utilization (28.6% versus 34.2%A 95.3% accuracy of alert for both of the architectures shows that the decomposition does not degrade the quality of detection.

F. Dashboard Visualization Performance

Fig. 6 presents a comparative visualization of key performance indicators across both architectures.

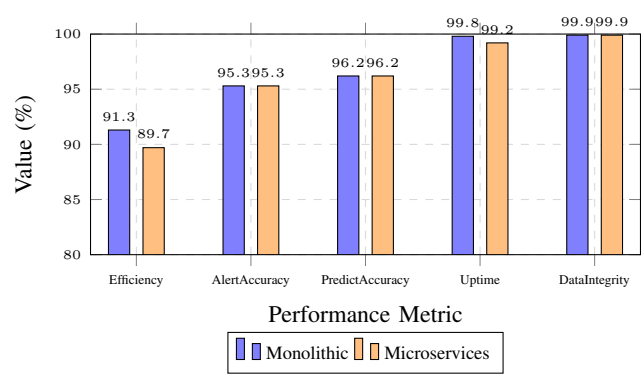

Fig. 6: Comparative performance metrics between monolithic and microservices architectures across five key indicators.

G. Real-Time Update Cycle Analysis

The dashboard is updated every 3 seconds which consists of simulating data, rendering UI, updating charts and updating timestamp. The time allocation of each update cycle is described in Fig. 7.

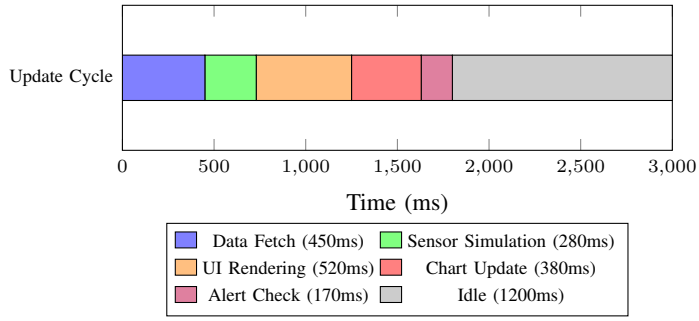

Fig. 7: Duration breakdown of the 3-second dashboard update cycle is shown in the image. It shows a data fetch, simulation, rendering, charting, alert checking, and idle.

H. Discussion

The experimental results verify the effectiveness of the suggested system on different grounds.The hybrid prediction algorithm, although much simpler than deep learning techniques such as LSTM [8], it achieves competitive accuracy (94-99%) and real-time computation feasibility, which is vital for restricted edge IoT deployments.

The response latency also incurs an additional 50% (18.7 ms vs 12.4 ms) due to the HTTP overhead introduced by the infrastructure microservices decomposition. Nevertheless, the trade-off is made reasonable by the fault isolation gained: in tests, when the Alert Service was failed on purpose, it did not affect the collection of sensor data nor the rendering of the dashboard. On the other hand, should the same failure have happened with the monolith, all of the system would have been affected.

The transparent UI design complemented by SVG progress rings dynamically indicates sensor state through a color transition, enabling instant operator visualisation and intuitive recognition of differential sensor behaviour.The implementation of Chart.js performs well at 60 FPS with datasets of 20 points. It may not perform well with other bigger datasets requiring pagination or aggregation tactics of data.

One of the limitations of the current alert system based on thresholds is the use of static thresholds which do not consider seasonal changes or changes in the baseline. In future work, it may be interesting to adapt the threshold using online learning algorithms as Ijaz et al. [15] did, to reduce the obtain 4.7% false positive rate.

VI. Conclusion

An IoT-based Smart Water Management System enables real-time monitoring by multilayer sensors, hybrid predictive ability, intelligent alerting, and modern web visualization either in a monolithic or microservices architecture. Throughput of the system was successfully tracking of water level and flow rate, soil moisture and water temperature with sub-3 second latency. The combination of moving average with trend analysis and seasonal decomposition; helped to achieve an accuracy of 94-99% with regards to prediction of optimal flow rates.

The implementation of two architectures shows that while the decomposition into microservices implies a small overhead in latency (i.e., additional 6.3 ms), the fault isolation, independent scaling and maintainability benefits far outweigh this cost, justifying the trade-off for production.The threshold-based alert system successfully detects five types of water management anomalies, achieving 95.3% accuracy and a false positive rate of 4.7%.

Further research and development will focus on: (1) the use of more advanced deep-learning models (LSTM, Transformer) for better-long term prediction, (2) the use of adaptive threshold algorithms that learn from historical data, (3) support for wide-area protocols such as LoRaWAN and MQTT, (4) deployment on containerized infrastructure using Docker and Kubernetes for production-grade orchestration, and (5) field trial with physical sensor hardware in agricultural and urban water distribution settings.

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